

Education

KTH Royal Institute of Technology, Machine Learning

Sweden
2023 – 2026

- *Thesis:* AI for Networked Robotics
- *Courses:* Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Robotics, Computer Graphics, Multi-Agent Systems, Statistical Methods

University of Surrey, Computer Science

UK
2019 – 2023

- *Thesis:* RoboKinesis
 - Controlling Robotic Arm with Computer Vision
- *Courses:* Programming, Algorithms, Software and Web Engineering, Artificial Intelligence, Deep Learning, Natural Language Processing

Experience

Computational Imaging and Vision Laboratory, National Institute of Informatics, Research Exchange Student

Japan
2025 – 2026
1 year

Video Motion Magnification

- Analyzed reflectance behavior of synthetic materials using BRDF parameters and Video Motion Magnification techniques, producing promising results that will inform future lab research. Developed Python implementations of signal processing algorithms, including Riesz Transform for motion/reflectance decomposition.
- Supervisor: [Dr. Imari Sato](#)

Ericsson Research, Master's Thesis Student

Sweden
2025 – 2025
1 year

Towards Adaptive Reinforcement Learning for Network-Aware Robotics via Quantization Techniques

- Applied adaptive RL with quantization to network-aware robot control, demonstrating improved model efficiency and policy interpretability on proprietary datasets. Extended the base algorithm with stability improvements and real-world control constraint alignment. Built end-to-end training and evaluation pipelines for network-constrained environments.
- Supervisor: [Dr. Christos N. Mavridis](#)

Visualization Course, KTH Royal Institute of Technology, Teaching Assistant (Visualization course)

Sweden
2024 – 2024
1 year

- Graded coursework and led support sessions, guiding students through core visualization concepts.

Centre for Vision, Speech and Signal Processing, University of Surrey, Summer Research Internship Student

UK
2023 – 2023
1 year

- Built a Gazebo simulation environment and a curriculum-based Reinforcement Learning pipeline using PyTorch. Trained a TurtleBot agent to autonomously score goals in simulation, supporting future deployment for RoboCup robotics competitions.
- Supervisor: [Dr. Oscar Mendez Maldonado](#)

Hawk-Eye Innovations, Machine Learning Engineer (Placement)

UK
2022 – 2023
1 year

- Improved UX and functionality of an internal React-based annotation tool used for training models. Developed C++ applications for data collection, visualization, and preprocessing used by tennis operators. Optimized large-scale data processing pipelines, significantly reducing preparation time for tennis model training. Prototyped ML pipeline improvements during hackathons, including evaluating YOLO-based pose-estimation models in PyTorch to explore potential accuracy and latency gains.

Publications

Genetically Modified Wolf Optimization with Stochastic Gradient Descent for Optimising Deep Neural Networks

Novel optimization approach combining metaheuristic algorithms and SGD to balance exploration and exploitation for training Deep Neural Networks.

Manuel Bradicic, Michal Sitarz, Felix Sylvest Olesen

arxiv.org/abs/2301.08950

Skills

Programming Languages

ML/AI Frameworks

Tools & Infrastructure

3D/Simulation

Languages

Polish

Native speaker

English

Fluent

Serbo-Croatian

Fluent